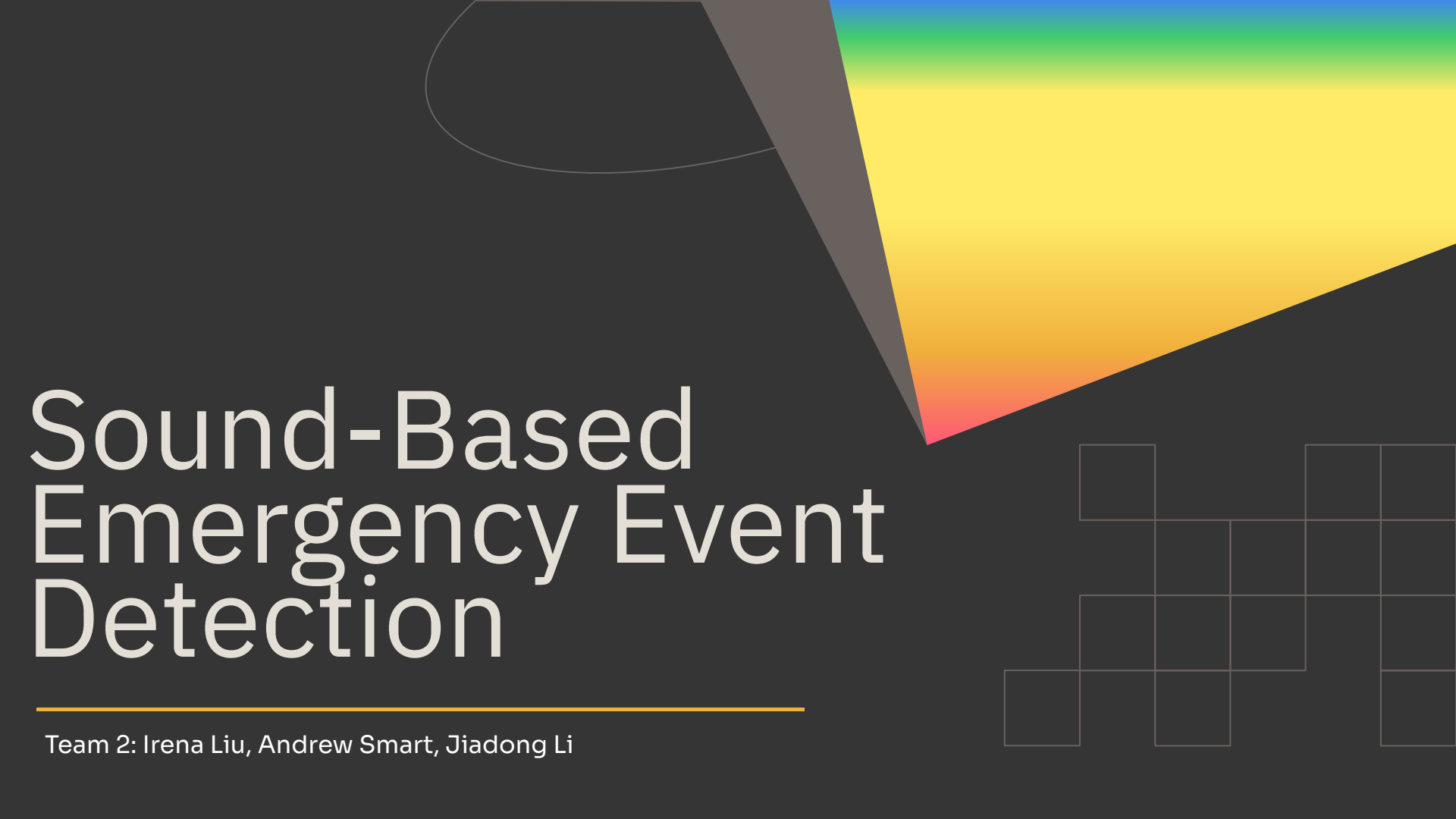


The background is a dark gray. In the top right, there is a large, abstract shape with a rainbow gradient from blue at the top to red at the bottom. A dark gray triangle points towards the bottom left of this shape. In the bottom right, there is a grid of white-outlined squares, some of which are missing, creating a fragmented pattern.

Sound-Based Emergency Event Detection

Team 2: Irena Liu, Andrew Smart, Jiadong Li

Motivation

A decorative background element consisting of a grid of squares in the top-left corner, with some squares shaded in a light gray color.

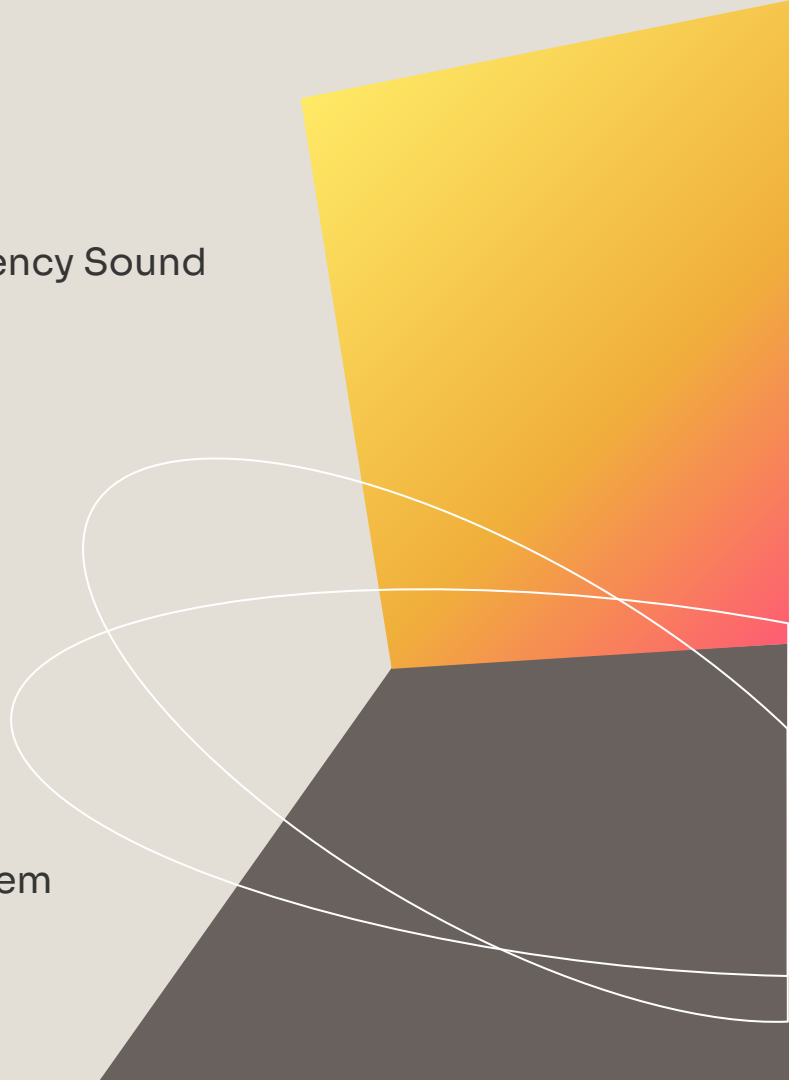
Main Idea

- A smart sensing system that recognize the Emergency Sound before user visualize them.

Possible Events

- Fire Alarm
- Siren
- Glass Breaking
- Screaming
- Loud Impact sounds
- Smoke Alarm

Why it Matters

- It helps in improving the smart home security system
 - Support the elderly people or people living alone
 - Useful for people with hearing difficulties
- 
- An abstract geometric design on the right side of the slide. It features a large yellow-to-orange gradient shape in the top right, a dark gray shape in the bottom right, and two white curved lines that sweep across the middle of the slide.

ML Component

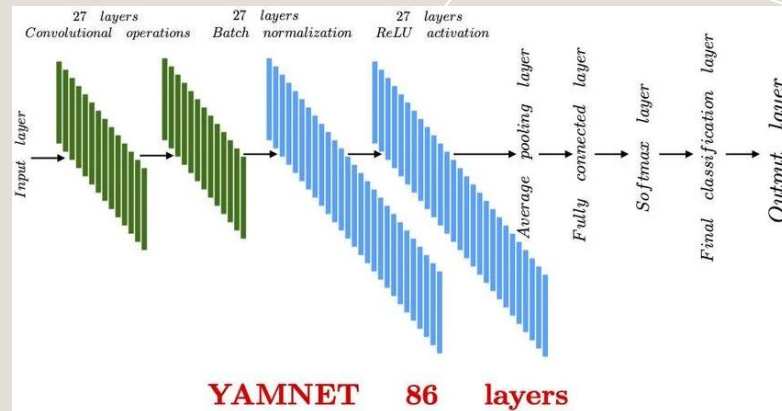
Task: Multi-class emergency sound classification

Pretrained Model: YAMNet

- The YAMNet is a pretrained audio event classification model trained on AudioSet

Possible Classes

- Fire Alarm / Smoke Alarm
- Siren
- Glass Breaking
- Normal Background Noise
- Scream



Model Design

Audio Waveform -> Pretrained YAMNet -> Audio Embeddings -> Classification Layers -> Labels

Sensing Component

Sensor Used: Microphone from laptop or phone

Input Data

Continuous audio stream, processed in 1 second windows

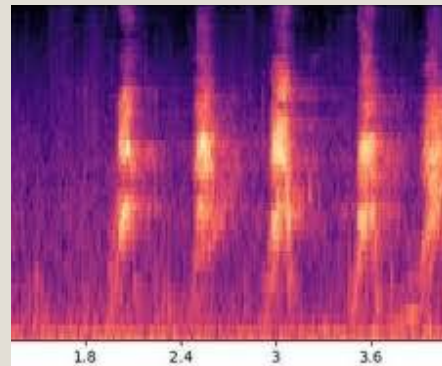
Captured at 16kHz sample rate

Preprocessing

Split audio into fixed-size overlapping windows

Convert each window into a mel spectrogram

Normalize input for consistent model performance

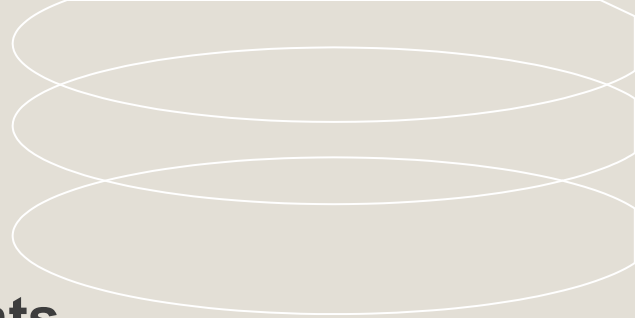


Microphone → Audio Windows → Mel Spectrogram → Model → Notification

Evaluation

Category	Metric	Success Criteria
Classification	F1-Score Mean of precision and recall	> 0.85 emergency classes
Timing	Detection Latency Processing time from sound to alert	< 1.5 seconds alert trigger
Robustness	False Alarm Rate % of non-emergency sounds flagged	< 2% normal background
Sensing	Sensitivity Effective range for clear recognition	Up to 10 meters detection

Live Demo Plan



Environment Setup

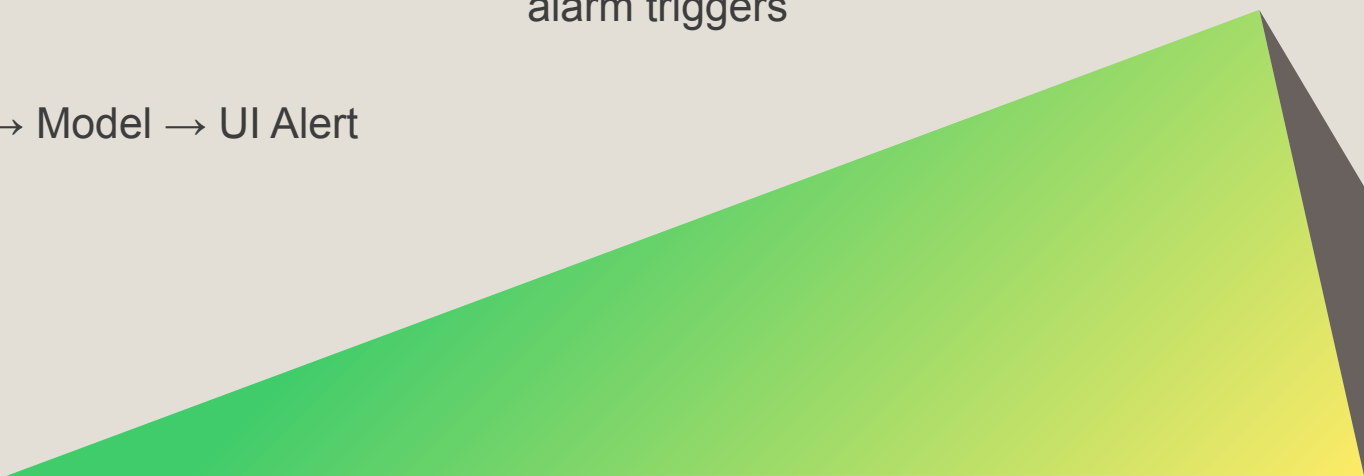
- Laptop running dashboard
- Sound sources, such as emergency smoke alarm and background noise

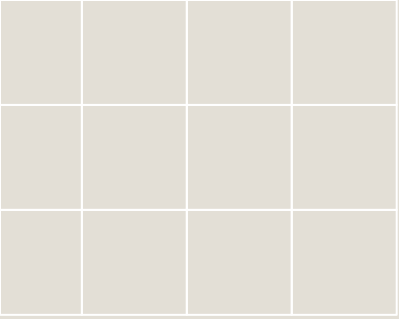
The Workflow

Mic → Mel-Spectrogram → Model → UI Alert

Components

- Real-time visualization of audio waveform
- Visual/Sound notification when alarm triggers





Any questions?

